

Clustered synapses develop in distinct dendritic domains in visual cortex before eye opening

Reviewed Preprint

v2 • June 13, 2024

Revised by authors

Reviewed Preprint

v1 • December 28, 2023

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Abstract

Synaptic inputs to cortical neurons are highly structured in adult sensory systems, such that neighboring synapses along dendrites are activated by similar stimuli. This organization of synaptic inputs, called synaptic clustering, is required for high-fidelity signal processing, and clustered synapses can already be observed before eye opening. However, how clustered inputs emerge during development is unknown. Here, we employed concurrent *in vivo* whole-cell patch clamp and dendritic calcium imaging to map spontaneous synaptic inputs to dendrites of layer 2/3 neurons in the mouse primary visual cortex during the second postnatal week until eye opening. We find that the number of functional synapses and the frequency of transmission events increase several fold during this developmental period. At the beginning of the second postnatal week, synapses assemble specifically in confined dendritic segments, whereas other segments are devoid of synapses. By the end of the second postnatal week, just before eye-opening, dendrites are almost entirely covered by domains of co-active synapses. Finally, co-activity with their neighbor synapses correlates with synaptic stabilization and potentiation. Thus, clustered synapses form in distinct functional domains presumably to equip dendrites with computational modules for high-capacity sensory processing when the eyes open.

eLife assessment

This **important** work provides insight into the activity and spatial organization of synapses during early postnatal development in the mouse visual cortex, using state-of-the-art tools to show that synapses are distributed in co-active clusters well before eye opening. The evidence supporting the claims is **convincing**, and this revised version provides additional methodological details about the experimental paradigm and image analysis. This work is of particular interest to the field of developmental neuroscience and can also be used by computational neuroscientists studying dendritic integration.

<https://doi.org/10.7554/eLife.93498.2.sa3>

Introduction

Building a functional brain requires neuronal circuits to be wired up with high specificity via synapses between connecting neurons. Developing neurons select their synaptic partners based on molecular cues such as adhesion molecules and subsequently refine their connections through activity-dependent synaptic plasticity, stabilization and elimination driven by spontaneous network activity and later, after the onset sensory function, by experience (Cline, 2003; Martini et al., 2021; Sanes and Zipursky, 2020). Electron microscopy revealed that excitatory synapses between neurons emerge during the end of the first postnatal week in mouse and rat sensory cortex (Blue and Parnavelas, 1983; De Felipe J. et al., 1997; Miller and Peters, 1981; Wildenberg et al., 2023). The addition of new synapses peaks at the end of the second postnatal week and synapse density in the primary visual and somatosensory cortices approaches adult levels once the eyes open at postnatal day (P) 14 (Blue and Parnavelas, 1983; De Felipe J. et al., 1997). While these structural studies clearly showed the importance of this developmental period for adding large numbers of synaptic connections in emerging circuits, how cortical dendrites establish functional inputs has not been investigated on the level of individual synapses *in vivo*.

In the adult cortex, synaptic inputs to dendrites are highly organized with sub-cellular specificity. Recent studies demonstrated that synaptic inputs of cortical pyramidal cells are clustered, such that synapses with similar activity patterns or stimulus selectivity are located near each other along their dendrites (Iacaruso et al., 2017; Ju et al., 2020; Kerlin et al., 2019; Otor et al., 2022; Scholl et al., 2021; Takahashi et al., 2012; Wilson et al., 2016; Winnubst et al., 2015). Since pyramidal cell dendrites integrate local inputs supra-linearly (Branco and Hausser, 2011; Harnett et al., 2012; Losonczy and Magee, 2006; Makara and Magee, 2013), this arrangement of clustered inputs is thought to allow for local synaptic integration in dendritic computational subunits (Larkum, 2022; Larkum and Nevian, 2008; Major et al., 2013; Tran-Van-Minh et al., 2015). Local dendritic integration dramatically increases the computational power of neurons (Poirazi and Mel, 2001). Furthermore, experimentally blocking supra-linear dendritic integration disturbs sensory stimulus sensitivity and selectivity in cortical neurons (Lavzin et al., 2012; Palmer et al., 2014; Smith et al., 2013; Xu et al., 2012). Finally, learning is associated with structural plasticity of clustered synapses (Cichon and Gan, 2015; Fu et al., 2012; Hedrick et al., 2022; Makino and Malinow, 2011; McBride et al., 2008). Together, these studies highlight the fundamental importance of the subcellular organization of synaptic inputs along dendrites.

This raises the question of how the precise subcellular organization of synapses is achieved during development. Synaptic clustering can be observed early on in the developing visual system of tadpoles and mice (Podgorski et al., 2021; Winnubst et al., 2015), suggesting that synaptic inputs develop with some degree of specificity, but the trajectory of functional synaptic input development has been unclear. Specifically, it is unknown (1) when synaptic inputs become clustered, (2) whether synapses are formed in a clustered state or arise in a random state to become sorted into clusters later, and (3) whether synapses with similar input patterns distribute smoothly along dendrites or emerge in distinct domains.

Here, we performed *in vivo* whole-cell patch clamp recordings with two-photon calcium imaging (Chen et al., 2011; Helmchen et al., 1999; Jia et al., 2010; Svoboda et al., 1997; Takahashi et al., 2012; Winnubst et al., 2015) of mouse primary visual cortex layer 2/3 neurons, allowing us to map functional synaptic inputs across dendritic arborizations in the course of the second postnatal week. We combined resonant scanning with piezo-driven focusing to image large proportions of a neuron's dendrites. The total dendritic length imaged per neuron was almost ten times larger than in our previous study (Winnubst et al., 2015), allowing us to map the large-scale distribution of synapses along developing dendrites.

We found that the number of functional synapses and the frequency of transmission events at individual synapses increase rapidly before eye opening. Furthermore, we observed that clustered synaptic inputs accumulate in spatially and functionally separated domains already at the beginning of the second postnatal week. By the end of the second postnatal week, dendrites had become almost entirely covered by functional domains of co-active neighbors, most likely through local plasticity driven by coincident spontaneous network activity. Thus, already before eye opening functional domains develop in cortical dendrites, which can serve as distinct computational modules for processing visual stimuli thereafter.

Results

Mapping synaptic inputs onto spine and shaft synapses

To visualize the activity and spatial organization of functional excitatory synapses in the developing mouse primary visual cortex, we imaged spontaneously occurring synaptic calcium transients in apical dendrites of pyramidal cells in L2/3 using the calcium indicator GCaMP6s in neonatal mice (**Fig 1A**). We combined resonant scanning and piezo-driven z-positioning to image a large area of the dendritic tree of individual neurons (**Fig 1B**). Calcium transients at individual synapses reflecting synaptic transmission (Jia et al., 2010; Takahashi et al., 2012; Winnubst et al., 2015) allowed us to map synaptic inputs across 12 dendritic areas from 11 mice at ages between postnatal day 8 to 13, the day before eye opening. Together, these dendrites hosted 354 functional synapses which produced 3440 synaptic transmission events. We recorded these neurons in voltage clamp configuration to measure barrages of synaptic currents arriving at the soma simultaneously with individual synaptic transmission events monitored by calcium imaging (**Fig 1C**).

In the mature cortex, most excitatory synapses of pyramidal neurons are located on spines (Berry and Nedivi, 2017; Kasai et al., 2021; Moyer and Zuo, 2018). Spines provide some chemical and electrical isolation from the dendrite's shaft and produce clearly detectable calcium signals evoked by presynaptic inputs (**Fig 2A**). However, during development, many cortical synapses are formed directly onto the shaft (Miller and Peters, 1981; Wildenberg et al., 2023) where synaptic calcium transients are masked by calcium influx triggered by back-propagating action potentials. Therefore, we prevented action potential firing by recording layer 2/3 pyramidal neurons in voltage-clamp mode during imaging and the intracellular sodium channel blocker QX314 in the patch-pipette. The holding potential was set to -30 mV to facilitate calcium influx through NMDA receptors (Jia et al., 2010; Takahashi et al., 2012; Winnubst et al., 2015). In this configuration we could image local calcium transients at both spine and shaft synapses (**Fig 2A**, B, see Methods for details and potential caveats) and map all functional excitatory synaptic inputs onto these dendrites across space and time (**Fig 2C**, Supplementary Figures 1 and 2).

Synaptic inputs increase rapidly during the second postnatal week

Previous electron microscopy studies revealed that synapses form at the highest rate during the second postnatal week in rodent sensory cortex (Blue and Parnavelas, 1983; De Felipe J. et al., 1997; Wildenberg et al., 2023); however, the course of functional synapse development *in vivo* has not been investigated yet. In line with these previous EM studies, we report here that the density of functional synapses increased significantly between P8 and P13 (**Fig 3A-B**). The frequency of synaptic transmission at individual synapses increased as well (**Fig 3C**). Hence, the number of synaptic transmission events in a given dendrite was higher by P12/13 ($n = 6$ dendrites) compared to the beginning of the second postnatal week (P8-10; $n = 6$ dendrites; **Fig 3D**). While the percentage of spine synapses was slightly higher in older dendrites, there was no significant

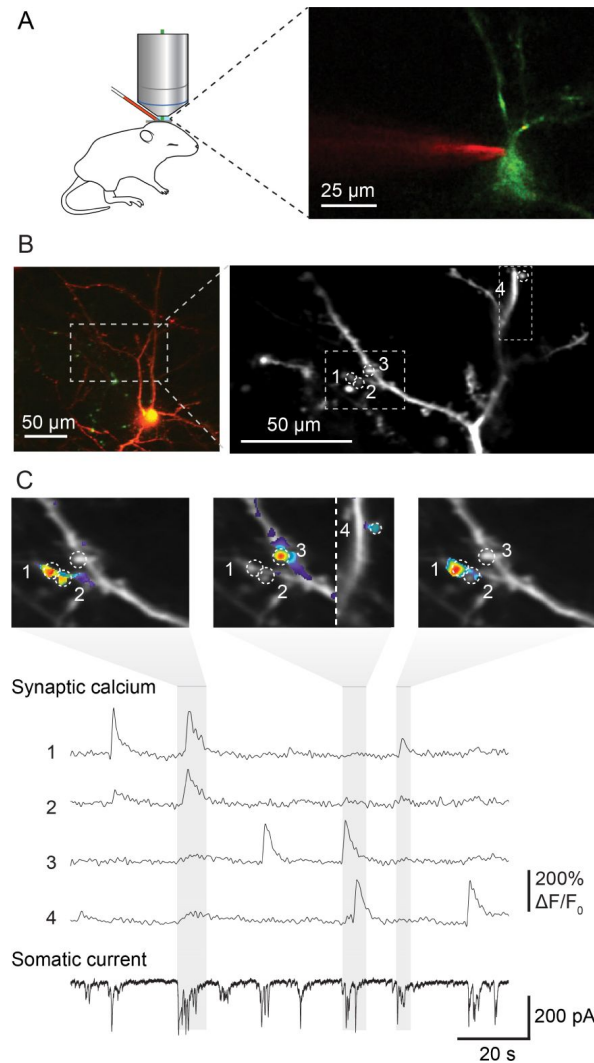


Fig. 1

Mapping functional synaptic inputs of V1 layer 2/3 pyramidal cell dendrites in vivo.

A. Schematic of experimental setup (left) and pyramidal cell in mouse visual cortex expressing DsRed and GCaMP-6s (right). Layer 2/3 neurons were targeted in vivo under two-photon guidance. Patch-clamp pipettes were coated with Alexa 594 for visualization.

B. Left: Layer 2/3 pyramidal neuron expressing DsRed and GCaMP-6s at postnatal day (P) 12 (D9 in Suppl. Figures 1 and 2). Right: High magnification view of the dendrite marked with dashed lines on the left. Dashed circles and numbers mark four individual synaptic sites. Dashed rectangle indicates views shown in C.

C. Spontaneous synaptic inputs were visible both as local increases in fluorescence at the four synaptic sites labeled in B, and as synaptic currents in the somatic whole-cell voltage-clamp recording. Grey vertical bars indicate three individual bursts, and panels above show synaptic calcium increases at the four example synapses during each labeled barrage. Synaptic transmission of individual synapses could be detected and distinguished clearly from that at neighboring synapses.

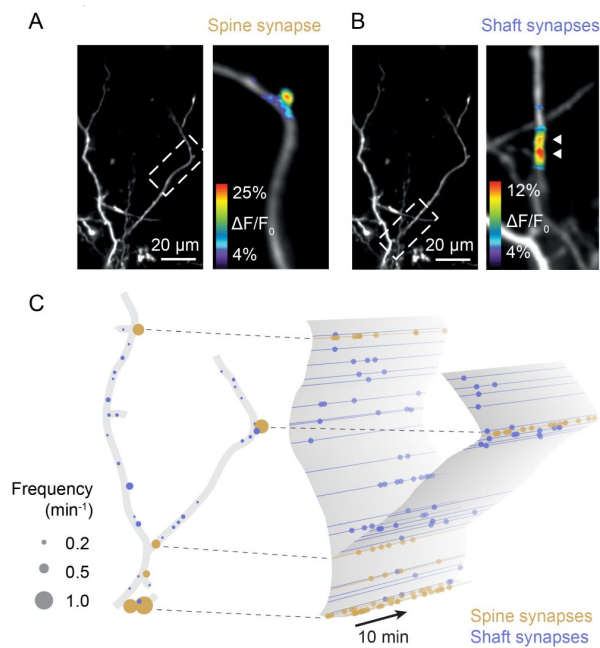


Fig. 2

Sampling synaptic transmission in spine and shaft synapses (D3).

A. Synaptic transmission in a spine of a layer 2/3 pyramidal neuron at P9.

B. Synaptic transmission at two neighboring shaft synapses of the neuron shown in A.

C. Schematic representation of the synaptic inputs of the neuron shown in A and B. Left: All functional synapses are labeled as individual discs. Yellow and blue discs represent spine and shaft synapses, respectively. The disc diameter indicates the observed transmission frequency at each synapse. Right: Schematic representation of synaptic transmission across time in the same dendrite. Dots represent individual synaptic transmission events.

relationship between age and the percentage of spine synapses across the sampled dendrites (**Fig 3E**). This is in line with previous observations that spine development is maximal after P14 (Summarized in: Lohmann and Kessels, 2014).

The distribution of synaptic transmission frequencies of all synapses showed that the majority of synapses received inputs at low frequencies, however, the long tail of the distribution demonstrated that a number of synapses were very active (**Fig. 3F**). In general, spine synapses were more active than shaft synapses in both young (P8-10; $0.55 \pm 0.09 \text{ min}^{-1}$ vs. $0.16 \pm 0.03 \text{ min}^{-1}$, mean $\pm$ SEM) and older animals (P12-13; $0.67 \pm 0.07 \text{ min}^{-1}$ vs. $0.23 \pm 0.03 \text{ min}^{-1}$; **Fig 3G**). While we detected synaptic transmission at the majority of spines, a fraction of structural spines remained inactive during our recordings in both younger ($25 \pm 20\%$) and older dendrites ($9 \pm 9\%$ for recording durations of 29 ± 8 and 26 ± 6 min, respectively; mean $\pm$ STD), possibly representing presynaptically silent synapses (Voronin and Cherubini, 2004).

Next, we investigated when individual synapses were activated in relation to the barrages of spontaneous synaptic inputs neurons receive during this developmental period (**Fig 4A**). Barrages were identified as transient inward currents of at least 10 pA, which consisted of multiple synaptic currents. We found that most synaptic inputs occurred in barrages (59% in young neurons, 61% in older neurons). The rate of barrages did not change significantly with age (**Fig 4B**); however, we did observe a trend towards an increase in synaptic charge transferred per barrage (**Fig 4C**), as well as a significant increase in the number of synaptic transmission events per barrage (**Fig 4D**) during development. As expected, the number of events observed per barrage correlated highly linearly with the charge transferred during each barrage (**Fig 4E**). In addition, the amplitude of individual synaptic transmission events was higher during larger barrages (**Fig 4F**), suggesting that multiple synaptic vesicles were released simultaneously during larger barrages.

Synapses are spatially organized along developing dendrites

In the mature mouse visual cortex, synaptic inputs are functionally clustered along layer 2/3 dendrites, such that neighboring synapses frequently have similar receptive fields (Iacaruso et al., 2017). During development, synaptic inputs are already clustered before eye-opening (Winnubst et al., 2015). Therefore, we explored here the developmental trajectory of dendritic input organization in general and synaptic clustering in particular. First, we noticed – mostly in young neurons – that some dendritic segments contained several synapses, whereas other segments were essentially devoid of active synapses (**Fig 5A**). To determine whether synapse distribution along the imaged dendrites was indeed patterned rather than random, we compared the distances between each synapse and its nearest distal and proximal neighbors with distances that were estimated after randomly shuffling synaptic locations along the dendrite. We found that the observed inter-synapse distance distribution differed significantly from randomized distributions in younger dendrites (P8-P10; **Fig. 5B**). In contrast, in older dendrites (P12-P13) synapse distribution was indistinguishable from randomized distributions. We concluded that in younger dendrites synapses accumulated in specific dendritic segments, and that with development the entire dendritic surface became covered by synapses more evenly.

Our finding that the distribution of transmission frequencies of individual synapses showed an extended tail (**Fig 3F**), made us wonder whether those synapses with high transmission frequencies showed any particular spatial distribution as well. We defined high-activity synapses as those in the top 20 activity percentile for each age. In line with our previous observation that spine synapses showed higher transmission rates than shaft synapses (**Fig 3G**) we found that the majority (76%) of high-activity synapses were located on spines. Furthermore, we observed that highly active synapses occurred frequently near each other (**Fig 5C**). Therefore, we tested whether there was a relationship between the activity level of a given synapse and the distance to its nearest high-activity neighbor. We discovered that synapses that were located within 10

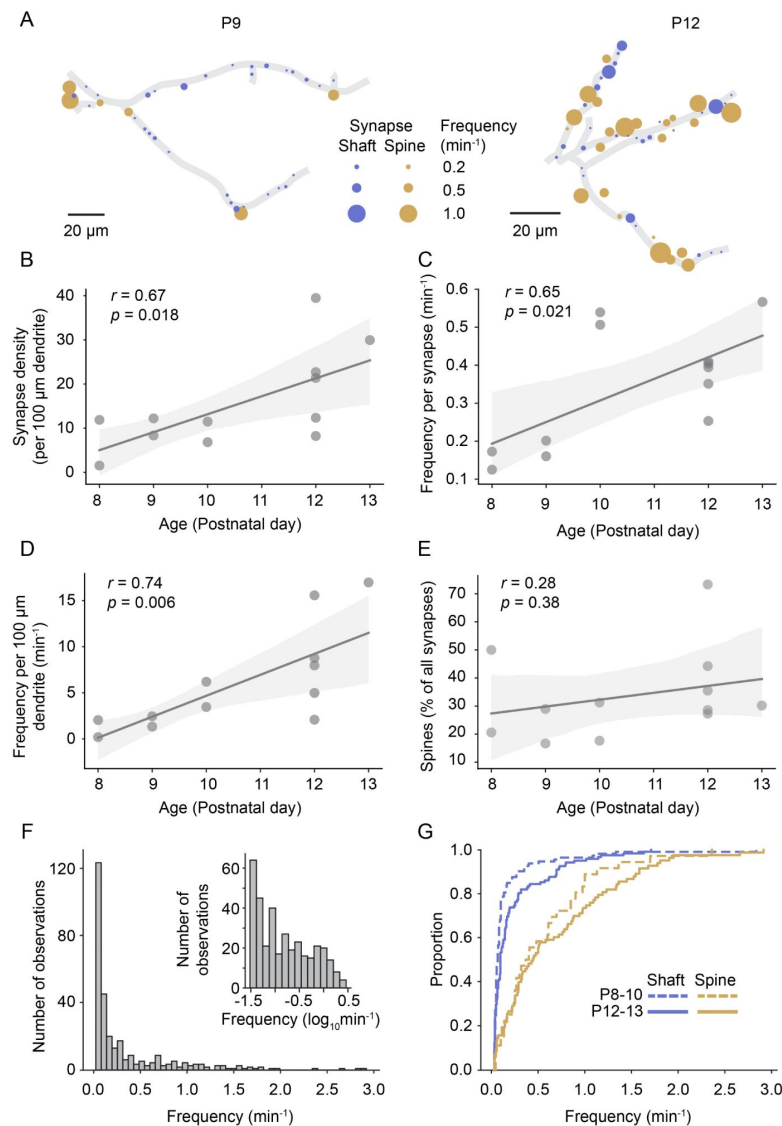


Fig. 3

Synapse density and synaptic transmission frequency increased rapidly during the second postnatal week.

A: Plots of layer 2/3 pyramidal cell dendrites from two experiments at P9 (D3) and P12 (D10), respectively. Each functional synapse is indicated as a disc where the diameter represents the frequency of transmission at shaft (blue) and spine (yellow) synapses.

B: The density of functional synapses increased with age. Each dot represents the number of synapses per 100 micrometer dendrite of one experiment. Statistical parameters in this panel and C-E are derived from linear regressions on the data from $n = 12$ experiments. The light grey area indicates the 95% confidence interval as in all following regression plots.

C: The frequency of synaptic transmission per synapse increased with age. Each dot represents the mean frequency across all synapses from one experiment.

D: The frequency of synaptic inputs along a stretch of dendrite increased multi-fold during the second postnatal week.

E: The percentage of synapses that are located on spines did not change significantly during the observed period.

F: The distribution of transmission rates across all sampled synapses. While most synapses show only low transmission frequency, the distribution is very long-tailed, demonstrating that few synapses show very high synaptic transmission rates. Inset: frequency distribution shown on a logarithmic scale.

G: Cumulative distribution of transmission frequencies across spine and shaft synapses in younger (P8-10) and older dendrites (P12-P13). The transmission frequency was higher at spine than at shaft synapses at both ages (each $p < 10^{-9}$, Mann-Whitney-U tests, P8-10: $n = 36$ spine / 113 shaft synapses, P12-13: $n = 83$ spine / 122 shaft synapses).

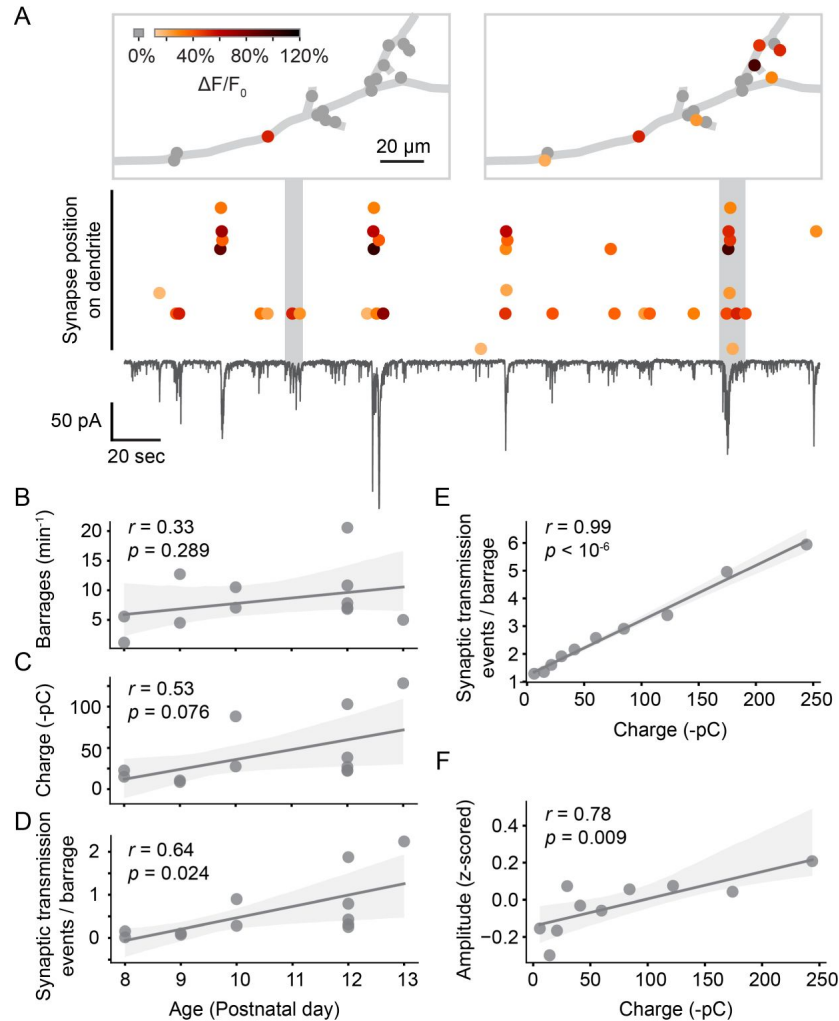


Fig. 4

Synaptic transmission at the synapse and soma level.

A: Synaptic transmission events (dots) at individual synapses and synaptic currents recorded simultaneously from the soma (black trace) in a layer 2/3 neuron at P10 (D6). Top: dendritic location of synapses that are active during two example barrages. Colors represent the amplitude of individual synaptic transmission events. Grey dots show synapses that were inactive during the respective barrages. Most synaptic transmission events occurred during barrages of somatic currents.

B: The number of barrages did not change significantly during the observed developmental period. Each dot represents the mean from one experiment. The light grey area indicates the 95% confidence interval as in all following regression plots.

C: The total charge transferred during barrages of synaptic currents showed a trend towards increases with age.

D: The number of synaptic transmission events observed during individual barrages increased with age.

E: The number of synaptic transmission events correlated linearly with the total charge transferred of barrages during which they occurred. Each data point shows the mean number of synaptic transmission events per barrage for geometrically binned barrage sizes.

F: The amplitude of synaptic transmission events at a given synapse correlated with the total charge transferred, suggesting that more than one vesicle was released per synapse during larger barrages. Each data point shows the mean amplitude (z-scored across all events of each synapse) per barrage for geometrically binned barrage sizes.

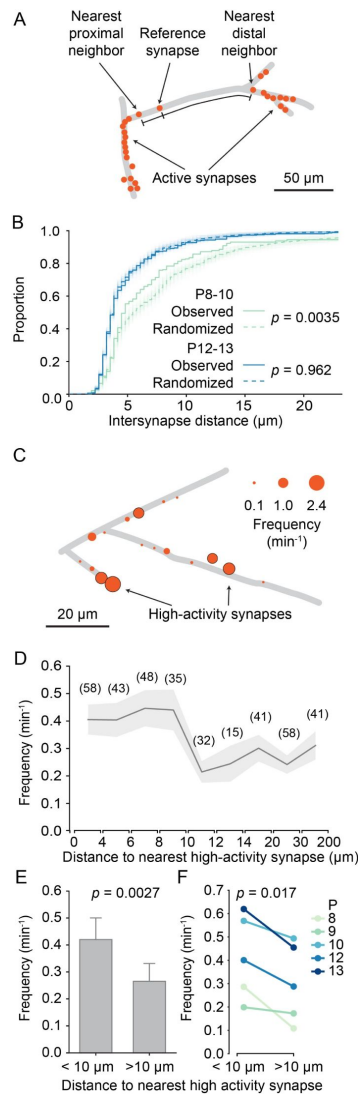


Fig. 5

Structured organization of synapses along developing dendrites.

A: Graphical representation of layer 2/3 pyramidal cell dendrites at P8 (D2). Each dot shows an active synapse. Several dendritic segments carried a high density of synapses; others did not receive functional synaptic inputs.

B: Cumulative distribution of inter-synapse distances between each synapse and their nearest proximal and distal neighbors. The inter-synapse distance distribution of synapses in younger dendrites (P8-10, green) differed significantly from randomized distributions (100 runs are shown as faint lines and their average as dashed line). In older dendrites (P12-13, blue), inter-synapse distributions did not differ from randomized distributions (Kolmogorov-Smirnov tests).

C: Graphical representation of layer 2/3 pyramidal cell dendrites at P10 (D5). The diameter of each disc represents the transmission frequency of a synapse. Synapses that were among the 20% most active synapse within their age group (high-activity synapses) are outlined in black. Frequently, highly active synapses were located in close proximity to other highly active synapses.

D: The mean transmission frequency of synapses was higher at synapses that were located nearby high-activity synapses. A clear drop in transmission frequency was observed at distances larger than 10 micrometers from the nearest high-activity synapse. In parentheses, the number of synapses averaged for each distance bin are shown. Grey area represents SEMs.

E: The mean transmission frequency was higher at synapses that were located within 10 micrometers of a high-activity synapse than at synapses farther away from high-activity synapses (Student's t-test for independent samples, 2-sided, $n = 181$ ($< 10 \mu\text{m}$) and 173 ($> 10 \mu\text{m}$) synapses. Error bars: SEM).

F: Within each age group, synapses that were located within 10 micrometers from an high-activity synapse were more active than those that were located farther away from high-activity synapses (Student's t-test for paired samples, 2-sided).

micrometers from a high-activity synapse were more active than synapses that were more distant to a high-activity synapse (**Fig 5D** [↗](#)). In fact, neighbors of high-activity synapses were significantly more active than those with larger distances from high-activity synapses (**Fig 5E** [↗](#)) and this relationship was highly consistent across all ages (**Fig 5F** [↗](#)).

Synaptic inputs are organized in functional domains

Having established that high-activity synapses were often surrounded by other high-activity synapses, we defined dendritic segments that contain high-activity synapses and their neighbors as dendritic domains (**Fig 6A** [↗](#)). Synapses that showed normal activity levels (i.e. non-high-activity synapses) were assigned to domains when they were 10 micrometer or less away from a high-activity synapse. High-activity synapses that were within 10 micrometers of each other were assigned to the same domain, whereas those that were farther away from each other were assigned to separate domains.

Assessing domain features across development revealed that their extent increased slightly (P8-10: $18.7 \pm 6.1 \mu\text{m}$; P12-13: $22.5 \pm 6.4 \mu\text{m}$). Moreover, both the number of synapses per domain (P8-10: 3.2 ± 1.7 ; P12-13: 6.0 ± 3.2) as well as the density of domains along dendrites (P8-10: 1.4 ± 0.7 ; P12-13: 2.6 ± 1.0 per 100 μm dendrite) approximately doubled during the second postnatal week (**Fig 6B-D** [↗](#)). The fraction of dendritic length covered by domains had increased significantly by the end of the second postnatal week (**Fig 6E** [↗](#); P8-10: $27.2 \pm 16.2\%$; P12-13: $61.3 \pm 29.4\%$, mean $\pm$ STD). Finally, the percentage of synapses within domains increased with age as well (P8-10: $46.2 \pm 7.4\%$, $n = 6$ dendrites; P12-13: $72 \pm 23.2\%$, $n = 6$ dendrites, mean $\pm$ STD; $p = 0.026$, t-test).

To relate dendritic domains (i.e. segments surrounding high-activity synapses) to synapse clusters (i.e. neighboring synapses with similar activity patterns), we compared local co-activity between synapses both inside and outside of dendritic domains. For each synapse, we determined a co-activity value by dividing the number of times this synapse was co-active with any synapse in the comparison group, e.g. its neighbors, by the number of times this synapse was active in total, normalized by the number of synapses in the comparison group. Next, we compared the local co-activity of synapses within domains to the local co-activity of synapses outside domains, and found that domain synapses showed much higher local co-activity in both younger (in: 0.08 ± 0.02 ; out: 0.02 ± 0.01 ; mean $\pm$ SEM) and older animals (**Fig 6F** [↗](#); in: 0.06 ± 0.01 ; out: 0.04 ± 0.01 ; mean $\pm$ SEM). Thus, synapses within a domain were more synchronized with their neighbors than synapses that were not part of a domain.

This finding raised the question whether dendritic domains are also functionally distinct from each other such that patterns of synaptic inputs received by one domain differed from the patterns received by the other domains on the same dendrite. To address this possibility, we quantified the co-activity of individual synapses with synapses from within their domain and compared that with their co-activity with synapses from other domains. We found that co-activity within domains was consistently higher than co-activity between synapses from different domains across all imaged dendrites (**Fig 6G-H** [↗](#)). These results demonstrate that synapses with similar input patterns emerge in distinct dendritic domains. Furthermore, they suggest that clustered synaptic inputs in the mouse visual cortex (Iacaruso et al., 2017 [↗](#); Wilson et al., 2016 [↗](#); Winnubst et al., 2015 [↗](#)) are confined to developmentally emerging distinct dendritic domains, and do not represent a smooth gradient of input patterns along the dendrite.

Local synaptic synchronicity correlates with synaptic activity changes

We then asked how uniform synaptic inputs became sorted into domains. Since we had discovered previously that local co-activity drives plasticity to cluster neighboring synapses (Niculescu et al., 2018 [↗](#); Winnubst et al., 2015 [↗](#)) and found here, that synapses within domains were more synchronized with their neighbors than synapses located outside domains (**Fig. 6F** [↗](#)), we

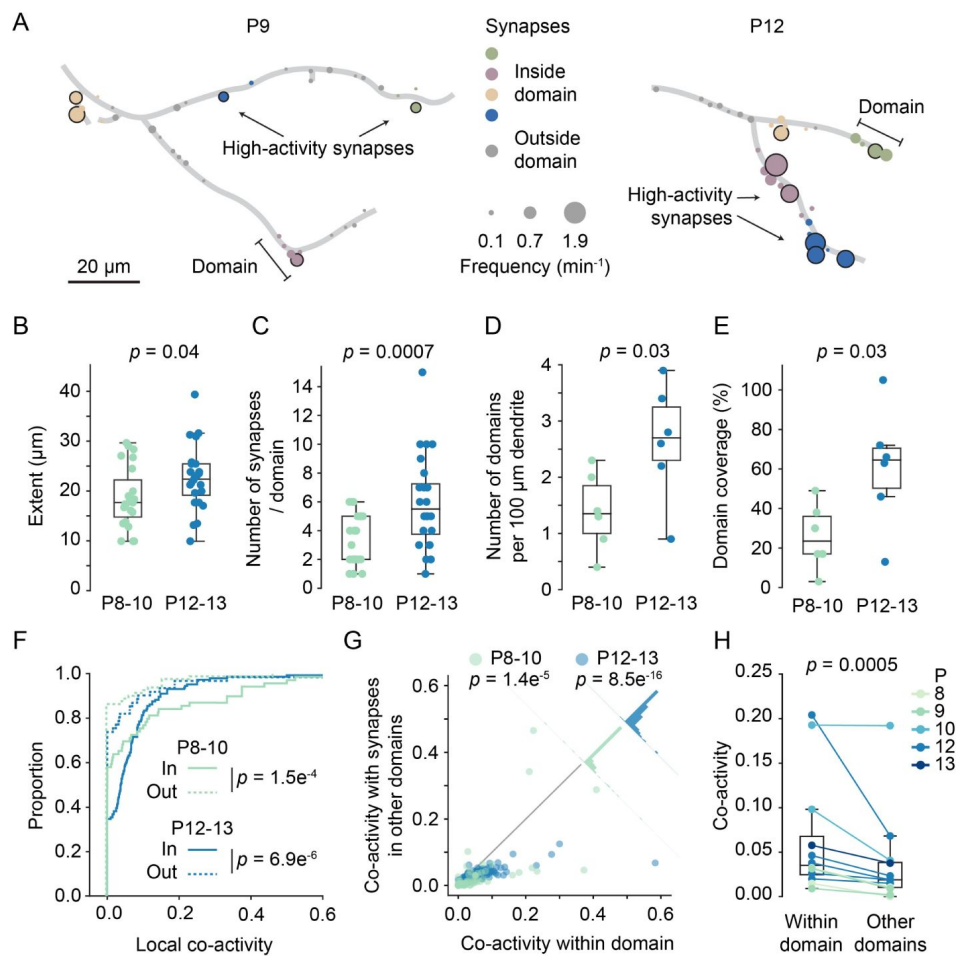


Fig: 6

Domains of clustered synapses emerged during the second postnatal week.

A: Graphical representation of layer 2/3 pyramidal cell dendrites at P9 (D3) and P12 (D7). Discs represent individual synapses. The disc size indicates synaptic transmission frequency, and high-activity synapses are labeled by black outlines. Individual domains are shown in different colors. Synapses located outside domains are shown in grey.

B, C: The extent of domains and the number of synapses increased during development. Each dot represents one domain (Student's t-test for independent samples, 2-sided, boxes indicate quartiles, and whiskers outline the entire distribution, except outliers).

D, E: the density of domains along dendrites and the fraction of dendrite covered by domains (E) increased with age. Each dot represents one dendrite (Student's t-test for independent samples, 2-sided, boxes indicate quartiles, and whiskers outline the entire distribution, except outliers).

F: Cumulative distributions of local co-activity values of all recorded synapses. The local co-activity of synapses inside domains was higher than that of outside-domain synapses (Man-Whitney-U test, 2-sided, $n = 213$ (In domain) and 173 (Outside domain) synapses. Error bars: SEM)

G: Synapses in domains were more co-active with their domain neighbors than with synapses in other domains. Each dot represents the co-activity of a synapse with its neighbors within the domain and its co-activity with synapses in other domains (Wilcoxon signed-rank test). The histograms perpendicular to the equality line show the distribution of differences.

H: For each experiment, the co-activity of synapses within domains was higher than their co-activity with synapses in other domains (Wilcoxon signed-rank test).

examined whether co-activity of neighboring synapses within domains predicted changes in their activity levels. To quantify changes in activity over time, we determined the Mann-Kendall score (Hussain and Mahmud, 2019) for each synapse. This score is positive for monotonic increases, negative for decreases, and zero in the absence of changes. Thus, synapses that increased in transmission frequency across recordings of an experiment yielded positive scores, those that underwent synaptic depression (i.e. decreased in transmission frequency) yielded negative scores, and those that maintained activity levels scored at zero (Fig 7A). We found that the Mann-Kendall score was more negative outside than inside domains, indicating that on average synapses outside domains tended to decrease in activity over time whereas those inside domains showed a more positive activity trajectory (Fig 7B). Next, we compared each synapse's Mann-Kendall score of synaptic plasticity with its local co-activity within 10 micrometers proximally and distally along the dendrite and revealed a significant positive correlation (Fig 7A, C), demonstrating that synapses that were synchronized with their neighbors became stabilized or more active compared to those that were out of sync with their neighbors.

Together, these findings showed that visual cortex neurons established functional synapses at a very high rate during the week leading up to eye opening and suggest that developing synapses became organized into functional, spatially distinct domains based on the synchronicity with their neighbors (Fig 7D).

Discussion

The fundamental idea that local processing and plasticity in dendrites is essential for high-level computations of the brain (Mel et al., 2017; Poirazi et al., 2003; Poirazi and Mel, 2001) has gained tremendous traction. Many recent studies show that synaptic inputs to excitatory neurons in the neocortex and hippocampus are clustered, such that synapses representing similar stimulus features are located near each other. However, when and how structured synaptic inputs assemble during development has been unclear. Here, we show that synaptic inputs to layer 2/3 pyramidal cells in the mouse visual cortex are structurally and functionally organized in distinct dendritic domains already during the second postnatal week, i.e. before eye opening and detailed visual input. Furthermore, we find that local co-activity within dendritic domains correlates with synaptic activity changes, indicating that synapses are sorted into distinct functional dendritic domains through plasticity mechanisms driven by spontaneous network activity (Fig 7D).

We obtained insight into the spatial organization of synaptic inputs onto layer 2/3 pyramidal cell dendrites by mapping synaptic transmission during spontaneously occurring network activity *in vivo*. Hence, the observed synaptic activity is the result of an interaction between the firing patterns of presynaptic neurons and the features of the individual synapses on the monitored neuron. Therefore, it is important to understand which properties of synaptic inputs we can infer from the recorded input patterns. Firstly, we observed that the distribution of synaptic transmission frequencies is very long-tailed: most synapses show only rare transmission events, whereas roughly 20 percent of synapses are highly active. In contrast, the largest population of input neurons (i.e. other layer 2/3 pyramidal cells) show very homogeneous firing activity during neonatal spontaneous network events where all neurons participate at a similar rate (Rocheffort et al., 2009; Siegel et al., 2012). Consequently, the distribution we observed here for individual synapses is most likely determined by release probability differences between individual presynaptic terminals along the dendrite. Furthermore, our current findings are congruent with our previous results which show that the transmission frequency of individual synapses is highly plastic and regulated over a large range by local synaptic interactions and a push-pull mechanism between mature BDNF and proBDNF in pyramidal neurons (Niculescu et al., 2018; Winnubst et al., 2015). Thus, transmission frequency is a synapse feature of large dynamic range that is most likely harnessed by the developing circuit to adjust connection strength in response to fine-scale activity patterns. This conclusion is supported by a previous study reporting that differences in

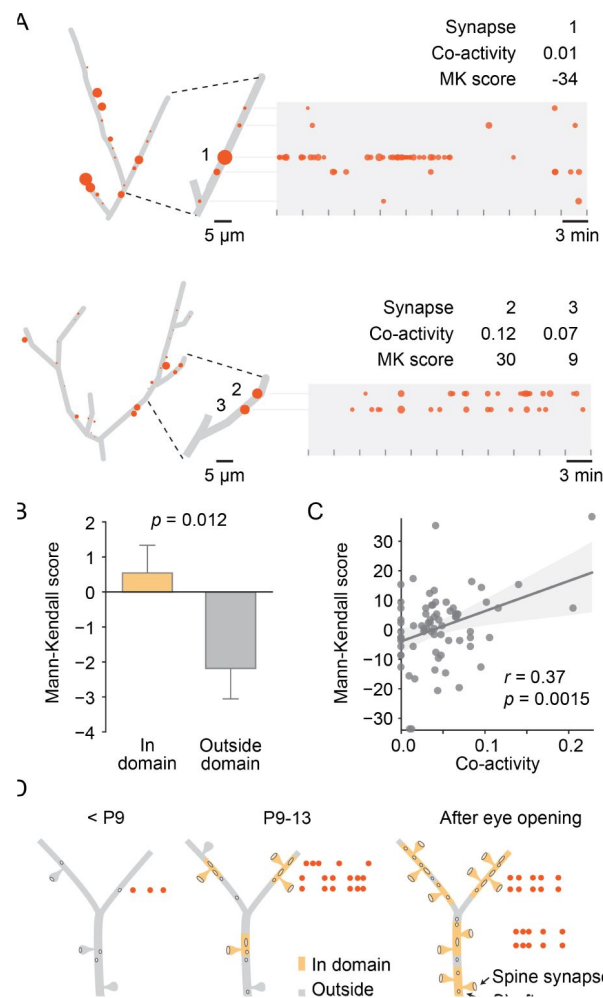


Fig. 7

Local co-activity of synapses inside dendritic domains correlated with synaptic potentiation.

A: Graphical representation of two layer 2/3 pyramidal cell dendrites at P12 (D5, D9). Left: Red discs on each dendrite represent active synapses, and their size shows relative activity levels. Activity of synapse (1) was mostly out-of-sync with its neighbors as indicated by a low co-activity score. This synapse was very active at the beginning of the experiment and stopped transmitting towards the end, resulting in a low Mann-Kendall (MK) score, which indicates a monotonic decrease in synaptic activity. Synapses (2) and (3) showed higher local co-activity and increased in activity over time as evidenced by a positive MK score. Tick marks indicate durations of individual 3 minute recordings. Red dots in the time plots represent individual synaptic transmission events. The size of each data point indicates the relative amplitude of each event.

B: The Mann-Kendall score, a measure of monotonic changes in synaptic transmission frequency, was higher for domain synapses than synapses located outside domains. (Student's t-test for independent samples, 2-sided, $n = 213$ (In domain) and 173 (Outside domain) synapses from all ages. Error bars: SEM)

C: The change in transmission frequency of a synapse was correlated with its local co-activity. Each dot represents one synapse with a transmission frequency of at least 0.6 min^{-1} (Linear regression).

D: Model for the establishment of domains of clustered synaptic inputs in layer 2/3 neurons of the developing visual cortex. Until P8, only a few active synapses are scattered across the developing dendrites. During the second postnatal week, synapses become sorted into distinct domains where in-sync synapses become stabilized or potentiated, and desynchronized synapses undergo synaptic depression. By eye-opening, most dendrites are covered by domains of synapses that are more co-active with each other than with synapses that are located in other domains.

cortico-cortical synaptic connections are mostly based on differences in synaptic release probabilities, and that these release probabilities change in response to altered input patterns (Markram et al., 1997 [↗](#)). Furthermore, the amplitude of calcium transients in axonal boutons varies over an order of magnitude, most likely representing a similar degree of variability in release probability, in juvenile rat layer 2/3 pyramidal cells (Koester and Sakmann, 2000 [↗](#)).

How many independent inputs, i.e. from unique presynaptic neurons, become connected to one dendritic domain? We observed a mean increase from three to six synapses per domain during the second postnatal week. In the adult, connected cortical neurons typically connect through fewer than ten synapses, most of which are located on different dendrites (Feldmeyer et al., 2006 [↗](#); Markram et al., 1997 [↗](#); Winfield et al., 1981 [↗](#)). However, axons that establish more than one synapse with individual dendritic segments have been observed as well. For example, in the mature somatosensory cortex, approximately ten percent of synapses are redundant, i.e. dendritic segments receive another synapse (or more) from the same axon (Kasthuri et al., 2015 [↗](#)). At the ages investigated in the present study, synapse density is lower than in the mature sensory cortex and redundant synapses most likely rarer. Nevertheless, our data set probably contains redundant synapses, too. Interestingly, full connectomic reconstruction of mature cortical dendrites and their presynaptic axons suggested that synaptic partner selection based on coincident activity may be responsible for the formation of multiple synapses of one axon with the same dendritic segment (Kasthuri et al., 2015 [↗](#)), indicating that multiple innervation may be generated through the here described plasticity mechanism that helps shaping domains of co-active synapses.

In the mature visual system, V1 pyramidal cell synaptic inputs are clustered, such that similar stimulus features are represented by neighboring synapses (Iacaruso et al., 2017 [↗](#); Wilson et al., 2016 [↗](#)). This input organization is thought to underlie local dendritic computations that maximize the computational power of cortical pyramidal neurons (Poirazi and Mel, 2001 [↗](#)). In fact, perturbing local dendritic integration has been shown to decrease the sensitivity or selectivity of cortical neurons to stimuli of different modalities (Lavzin et al., 2012 [↗](#); Palmer et al., 2014 [↗](#); Smith et al., 2013 [↗](#); Xu et al., 2012 [↗](#)). Which stimulus features may be encoded by synaptic domains established before eye opening? In the adult, neighboring synapses are more likely to share receptive field properties than more distant synapses. For example, in ferret V1 layer 2/3 neurons, synapses are clustered according to orientation preference (Wilson et al., 2016 [↗](#)), whereas in mouse V1 layer 2/3 neurons, synapses are clustered based on receptive field localization within the visual field, as well as receptive field similarity, but not on orientation preference (Iacaruso et al., 2017 [↗](#)). Thus, different sensory input features can be arranged in neighborhoods along dendrites, but exactly which features are clustered differs between species. In the retina, spontaneously generated activity travels in waves with specific directions as well as wavefronts of defined orientations. In addition, neighboring neurons are frequently co-active during wave propagating (Feller et al., 1997 [↗](#); Meister et al., 1991 [↗](#)). Hence, retinal waves encode retinotopy as well as direction and orientation selectivity. Since wave activity in retinal ganglion cells is transmitted along the central visual pathways from the retina to the visual cortex (Ackman et al., 2012 [↗](#); Hanganu et al., 2006 [↗](#); Siegel et al., 2012 [↗](#)), retinal waves can shape dendritic domains based on neighborhood relationships of retinal ganglion cells representing location in visual space, and orientation and direction selectivity (Kirchner and Georgieva, 2021 [↗](#)). Finally, our observation that functional synapses form in distinct domains along dendrites, whereas other dendritic segments are devoid of synapses at the beginning of the second postnatal week, indicates that synapses cluster in defined domains rather than distributing evenly along dendrites.

Together, patterned spontaneous activity can be sufficient to sort synaptic inputs into domains to establish discrete computational modules along dendrites to prepare the visual cortex for processing with high sensitivity and selectivity before the eyes open. This “first guess” will be refined further by visual inputs after eye opening, based on the statistics of sensory inputs in the actual environment.

Methods

Animals

All experimental procedures were approved by the Institutional Animal Care and Use Committee of the Royal Netherlands Academy of Arts and Sciences. We used 11 C57BL/6J mouse pups between P8 and P13.

Plasmids

For *in utero* electroporation, GCaMP6s (Addgene plasmid 40753; Douglas Kim; Chen et al., 2013 [\[1\]](#)) was cloned into pCAGGS and used in combination with DsRed Express in pCAGGS (Winnubst et al., 2015 [\[2\]](#), gift from Christiaan Levelt). Cells co-expressed the fluorescent protein DsRed for somatic targeting and structural information.

In Utero Electroporation

Constructs were introduced through *in utero* electroporation at E16.5. Pyramidal neurons in layer 2/3 of the visual cortex were transfected with GCaMP6s (2 mg/ml) and DsRed (0.5–2 mg/ml) at E16.5 using *in utero* electroporation (Winnubst et al., 2015 [\[2\]](#)). Pregnant mice were anesthetized with isoflurane and a small incision (1.5–2 cm) was made in the abdominal wall. The uterine horns were removed from the abdomen, and DNA (1 μ l) was injected into the lateral ventricle of embryos using a sharp glass pipette. Voltage pulses (five square wave pulses, 30 V, 50 ms duration, 950 ms interval, custom-built electroporator) were delivered across the brain with tweezer electrodes covered in conductive gel. Uterine horns were rinsed with warm saline solution and carefully returned to the abdomen, after which the muscle and skin were sutured.

In vivo electrophysiology and calcium imaging

The surgery and stabilization for the *in vivo* calcium imaging experiments were performed as described previously (Siegel et al., 2012 [\[3\]](#); Winnubst et al., 2015 [\[2\]](#)). Animals were anesthetized with 2% isoflurane, which was reduced to 0.7%–1% after surgery. We previously reported that although this low level of anesthesia does reduce the frequency of spontaneous network events, relative to that seen in awake animals, it does not change the basic properties of spontaneous network activity, such as participation rates and event amplitudes (Siegel et al., 2012 [\[3\]](#)). Furthermore, we found previously that synaptic inputs are organized very similarly *in vivo* under light anesthesia and in slice cultures (Winnubst et al., 2015 [\[2\]](#)). Together, these observations indicate that the activity patterns investigated here are not or only slightly affected by low-level anesthesia. Glass electrodes (4.5–6 M Ω) were fluorescently coated with BSA-Alexa 594 to allow targeted whole-cell recordings (Sasaki et al., 2012 [\[4\]](#)). To visualize synaptic inputs, action potentials were blocked with QX314 in the intracellular solution (120 mM CsMeSO₃, 8 mM NaCl, 15 mM CsCl₂, 10 mM TEA-Cl, 10 mM HEPES, 5 mM QX-314 bromide, 4 mM MgATP, and 0.3 mM Na-GTP; Takahashi et al., 2012 [\[5\]](#); Winnubst et al., 2015 [\[2\]](#)). Currents were recorded in voltage clamp mode at 10 kHz and filtered at 3 kHz (Multiclamp 700b; Molecular Devices). To facilitate the detection of synaptic calcium transients, neurons were depolarized to –30 mV to increase NMDA receptor activation (Takahashi et al., 2012 [\[5\]](#); Winnubst et al., 2015 [\[2\]](#)). When switching to –30 mV, the baseline fluorescence of the patched neurons increased, but not of the neighboring neurons, allowing unequivocal identification of all dendrites in the imaged volume that belonged to the patched neuron (Fig S3). No correction was made for the liquid junction potential. Barrages were identified as fast rising inward currents of at least 10 pA, followed by a return to baseline, which consisted of multiple synaptic currents and were separated by at least three seconds.

Image acquisition

In vivo calcium imaging was performed on a Nikon (A1R-MP) with a 0.8/16x water-immersion objective and a Ti:Sapphire laser (Chameleon II, Coherent). Dendrites were imaged at 5–15 Hz with a pixel size of 0.32 μm in single planes or small stacks with planes spaced 1.5 – 2 μm . We recorded the movement signal of the scan mirrors to synchronize calcium imaging and electrophysiology.

Image processing

To remove drift and movement artifacts from each recording, we performed drift correction using NoRMCorre (Pnevmatikakis and Giovannucci, 2017 [DOI](#)). Each recording was aligned to the first recording in the series to remove any movements between recording sessions. Delta-F stacks were made using the average fluorescence per pixel as baseline. Local synaptic calcium transients were clearly detectable in the delta-F stacks.

Regions of interest (ROIs), representing putative synapses, were hand-drawn using ImageJ (NIH) at sites where spines were visible and at clearly observed activity sites (Fig. S4). Finally, all remaining dendritic segments were filled with ROIs to ensure that no sites were missed. Putative synaptic transmission events were automatically identified across all ROIs as mean fluorescence increases that were at least two-fold higher than the noise level. These putative signals were then manually reviewed using the full image recordings. Signals were rejected, if they were shorter than two frames or coincided with movement. Furthermore, we ensured that there was a clearly visible local maximum at the location of the synapse and that it was stable for the duration of the signal. Automated transient detection and further data processing was performed using custom-made Matlab (MathWorks) and Python software (Python Software Foundation).

Identification of spine and shaft synapses

We identified spine and shaft synapses as follows. Synaptic calcium transients that occurred in small appendages from the dendrite were defined as transmission events at spine synapses (e.g. **Fig 2A** [DOI](#) and Fig S5A). Synaptic calcium transients located in the center of the dendrite, but at sites that showed small, sharply confined areas of increased fluorescence and where 3D reconstructions suggested that a spine may be located above or below the dendrite in the z-dimension were defined as spine transmission events as well (e.g. S5B). Synaptic calcium transients that occurred in the center of “smooth” dendritic stretches (without clearly defined spots of increased brightness) were identified as transmission events at shaft synapses (e.g. **Fig 2B** [DOI](#) and Fig S5A). Since the z-resolution of 2-photon microscopy is lower in the z-dimension than in xy, we may have misidentified spine synapses as shaft synapses in a few cases.

Statistical Analysis

One dendrite was imaged from each animal except once, when two dendrites were imaged in a single cell in a P12 animal. We used only synaptic sites that were active with a transmission frequency of at least 0.03 per minute. We tested whether the localization of functional synapses was structured or random along the dendrite. To establish randomized distributions of synapse positions, we first assigned “empty” positions between the positions of actual functional synapses such that the entire imaged dendrite was covered with real or empty positions at the typical minimal inter-synapse distance ($4 \pm 1 \mu\text{m}$). Next we determined all pair-wise distances between all synapses and all empty locations along dendrites using the Shortest Path with Obstacle Avoidance function of Matlab. Then we shuffled all functional synapses across the real and empty positions randomly. For each shuffle and the observed localizations, we determined the distances of each synapse to its nearest proximal and distal neighbors. Finally we compared the results of 100 shuffles with the observed distribution using a Kolmogorov Smirnov test.

The co-activity of a given synapse (e.g. with its neighbors within 10 micrometers, with its neighbors in the same domain or with synapses in other domains) was determined by dividing the number of observed transmission events at the observed synapse, by the number of transmission events of the comparison group (e.g. its neighbors) during the same barrages when the observed synapse was active. This number was divided by the number of synapses in the comparison group to normalize to the size of the comparison group. We defined co-activity based on co-transmission during the same barrage, because we had previously discovered that co-activity during barrages is crucial for local synaptic plasticity (Winnubst et al., 2015 [↗](#)).

Domains were defined as dendritic segments that contained at least one high-activity synapse (see Results section for definition). If the distance between two high-activity synapses was ten micrometers or less, both were assigned to the same domain. Normal synapses (those that were not high-activity synapses) were assigned to a domain, if their distance to the nearest high-activity synapse was ten micrometers or less. Normal synapses located between two high-activity synapses from distinct domains with 20 micrometer or less distance were assigned to the domain of the closer high-activity synapses. The extent of dendritic domains was determined as the distance between its most distant members. The length was padded at the edges with five micrometers from a high-activity synapse, if there was no normal member within this distance. Thus single high-activity synapse domains had a length of ten micrometers.

For normally distributed data and data with relatively low sample numbers (6-8), where normality tests are underpowered, we used paired and unpaired, 2-tailed Student's t-tests as recommended (de Winter, 2013 [↗](#)). For data that were not normally distributed, we used non-parametric statistical comparisons, Man-Whitney-U tests for independent samples and Wilcoxon signed-rank tests for paired data. To test whether investigated parameters changed with age, we performed linear regression analyses. To test the differences between distributions, we used a Kolmogorov-Smirnov test.

Author contributions

Conceptualization: A.H.L., J.E.C. and C.L.; Investigation: A.H.L., J.E.C.; Analysis: A.H.L., J.E.C. and C.L.; Writing: A.H.L., J.E.C. and C.L..

Acknowledgements

We thank Jan Kirchner and Julijana Gjorgjieva for discussions, Christiaan Levelt for sharing plasmids, and Helmut Kessels, Alexander Heimel, Wei Wei, David Cabrera, Tamara Buijs, and Julijana Gjorgjieva for critically reading this manuscript. This work was supported by grants of the Netherlands Organization for Scientific Research (NWO, ALW Open Program grants, no. 819.02.017, 822.02.006 and ALWOP.216; ENW Open Competition grant no. OCENW.KLEIN.535, ALW Vici, no. 865.12.001), ZonMW (Top grant no. 9126021) and the “Stichting Vrienden van het Herseninstituut” (all CL.).

Supplementary Figures

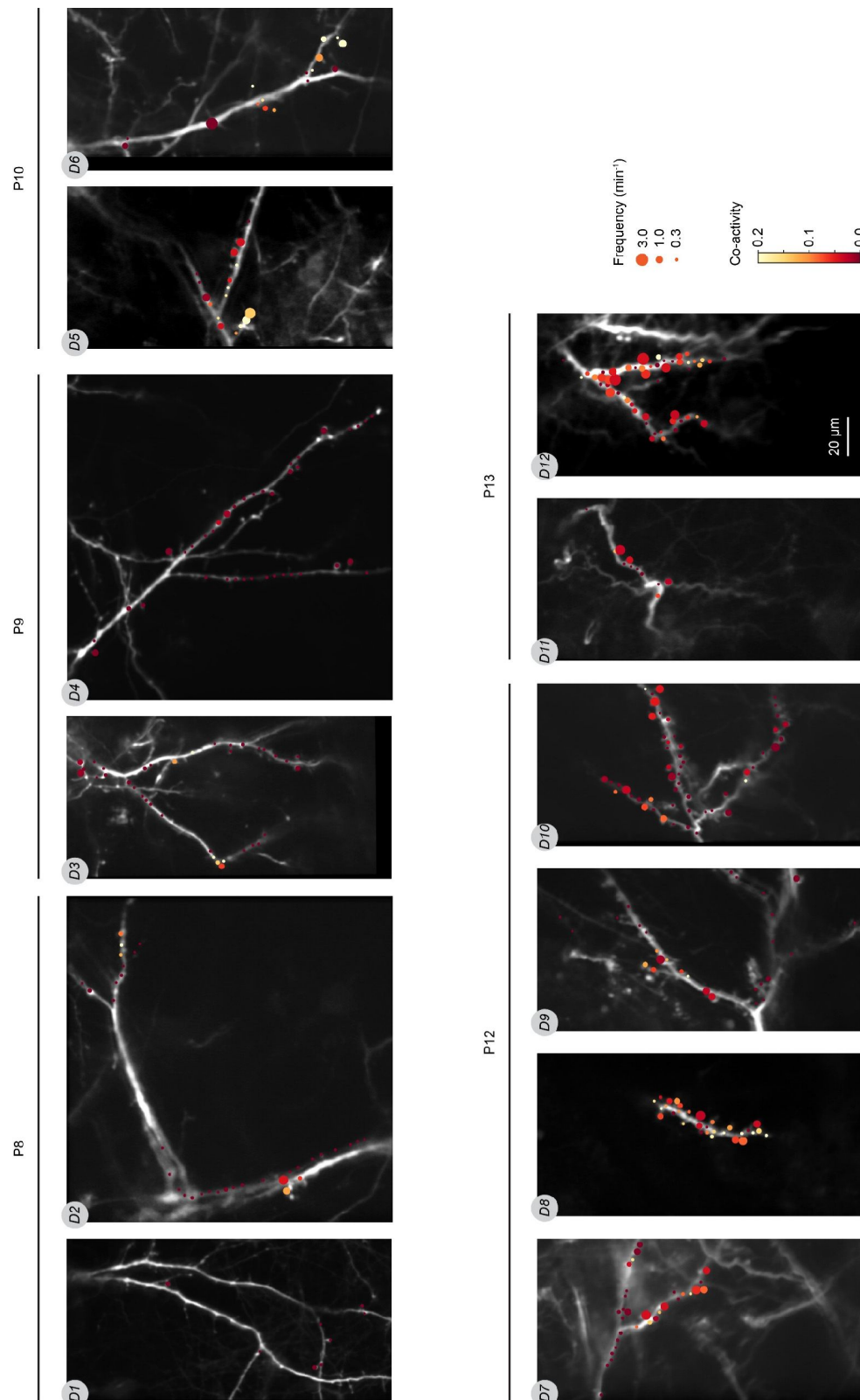


Fig. S1

Overview of all imaged dendrites and synapses.

Each synapse is represented as a disc whose size and color indicate transmission frequency and local co-activity, respectively. D1-D12 are each dendrite's identifier. The scale bar applies to all images.

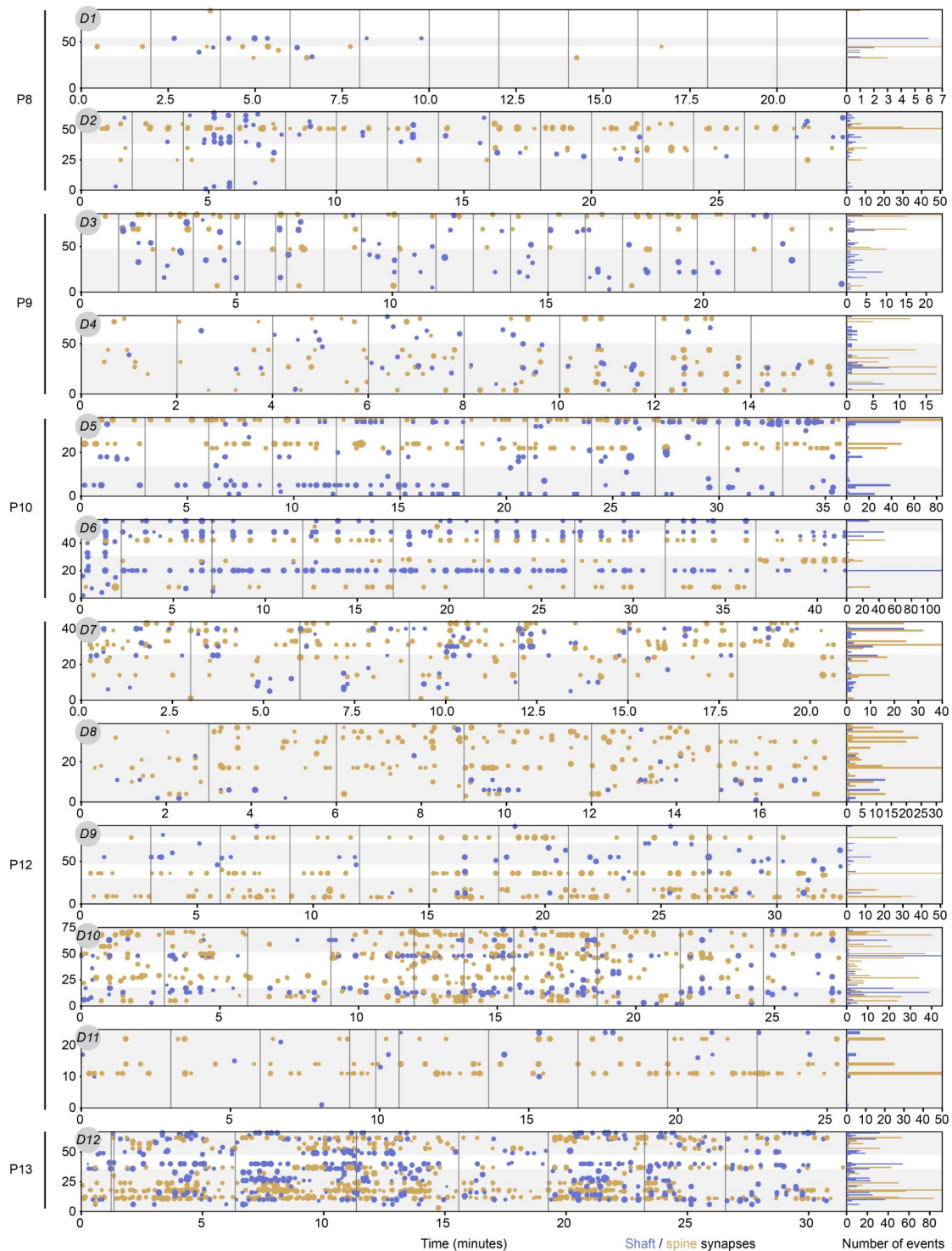


Fig. S2

Overview of all recorded synaptic transmission events across all imaged dendrites.

Y-axes show synaptic position along dendrites. White and grey shaded areas represent individual dendritic segments. Each synaptic transmission event is shown as a disc in yellow (spines) or blue (shaft synapses). Disc size indicates the relative amplitude of each event (z-scored for each synapse). Bar plots on the right show the number of transmission events detected at each synapse. Individual dendrites are shown in the same sequence as in Fig. S1.

Fig. S3

Identification of the patched neuron and its dendrites

Composite of the DSred channel (grey) and the GCaMP6s channel (green) after switching the holding potential to -30 mV of a layer 2/3 pyramidal neuron at P8 (D1). The depolarization triggers a transient calcium increase in the patched neuron, which labels its dendrites (green) clearly above the background (grey processes).

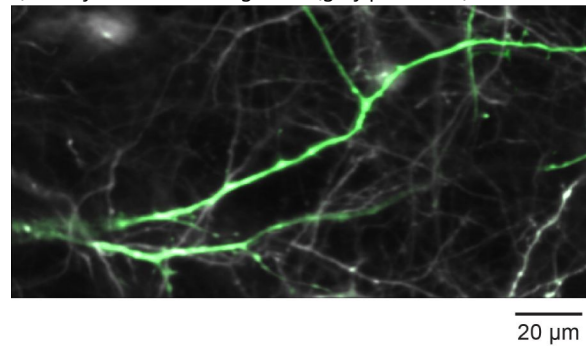


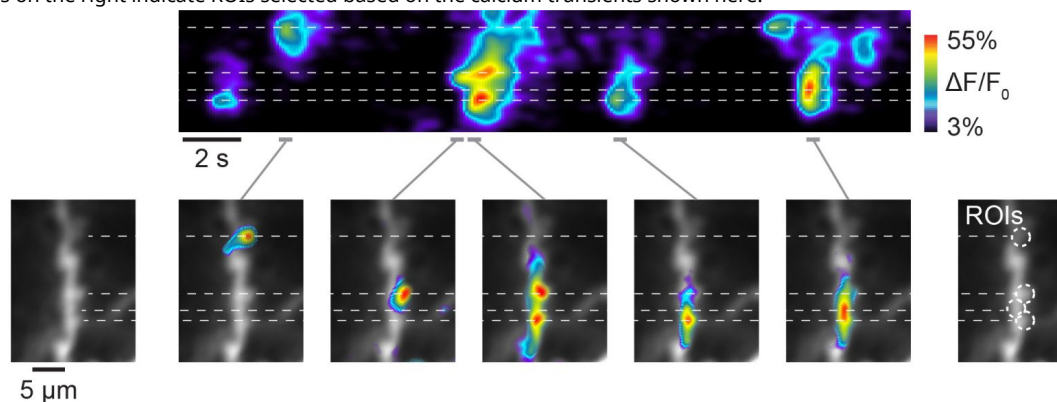
Fig. S4

ROI selection.

Synaptic calcium transients in a dendrite of a layer 2/3 pyramidal neuron at P13 (D12).

Top: occurrence of synaptic calcium transients in the dendritic stretch shown on the left. Stippled lines indicate the position of several transients. Bottom: Spatial representation of the events shown above at the time points indicated by the grey bars.

Circles on the right indicate ROIs selected based on the calcium transients shown here.



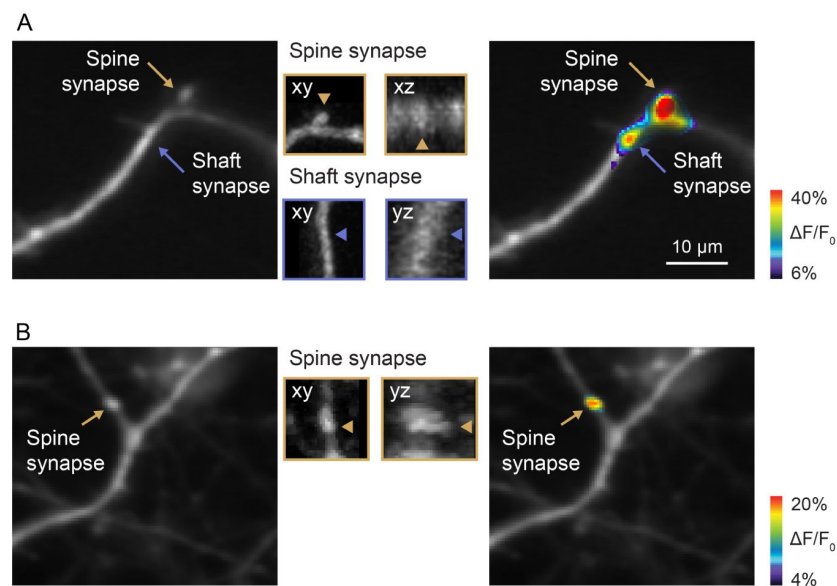


Fig. S5

Identification of shaft and spine synapses.

A: Dendrite of a layer 2/3 pyramidal neuron at P9 (D3). Left: Arrows indicate the positions of a spine (yellow) and a shaft synapse (blue). Middle: xy- and xz/yz-views show the dendritic segments around these synapses acquired in a z-stack before the here shown time-lapse recording. Arrow heads point to the same synapses as in the left panel. Right: Synaptic transmission events at these synapses.

B: Dendrite of a layer 2/3 pyramidal neuron at P8 (D1). The arrow points to a putative spine synapse located in the axis of the dendritic shaft. xy- and yz-views show the dendritic segment around this synapse acquired in a z-stack before the here shown time-lapse recording. The arrowheads points to a putative spine below the dendrite in the image plane. Right: synaptic transmission event at this synapse.

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Editors

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Reviewer #1 (Public Review):

Summary:

Using concurrent in vivo whole-cell patch clamp and dendritic calcium imaging, the authors characterized how functional synaptic inputs across dendritic arborizations of mouse primary visual cortex layer 2/3 neurons emerge during the second postnatal week. They were able to identify spatially and functionally separated domains of clustered synapses in these neurons even before eye-opening and characterize how the clustering changes from P8 to P13.

Strengths:

The work is technically challenging and the findings are novel. The results support previous EM and immunostaining studies but really provide in vivo evidence on the time course and the trajectory of how functional synaptic input develop.

Weaknesses:

The authors have provided additional details about the analyses and have adequately addressed all my concerns.

<https://doi.org/10.7554/eLife.93498.2.sa2>

Reviewer #2 (Public Review):

In this study, Leighton et al performed remarkable experiments by combining in-vivo patch-clamp recording with two-photon dendritic Ca²⁺ imaging. The voltage-clamp mode is a major improvement over the pioneer versions of this combinatorial experiment that had led to major breakthroughs in the neuroscience field for visualizing and understanding synaptic input activities in single cells in-vivo (sharp electrodes: Svoboda et al, Nature 1997, Helmchen et al, Nature Neurosci 1999; whole-cell current-clamp: Jia et al, Nature 2010, Chen et al, Nature 2011. I suggest that these papers would be cited). This is because in voltage-clamp mode, despite a full control of membrane voltage in-vivo is not realistic, is nevertheless most effective in preventing back-propagation action potentials, which would severely confound the measurement of individual synaptically-induced Ca²⁺ influx events. Furthermore, clamping the cell body at a strongly depolarized potential (here the authors did -30mV) also facilitates the detection of synaptically-induced Ca²⁺ influx. As a result, the authors successfully recorded high-quality Ca²⁺ imaging data that can be used for precise analysis. To date, even in view of the rapid progress of voltage-sensitive indicators and relevant imaging technologies in the recent years, this very old 'art' of combining single-cell electrophysiology and two-photon imaging (ordinary, raster-scanned, video-rate imaging) of Ca²⁺ signals still enable measurements of the best-level precision.

On the other hand, the interpretation of data in this study is a bit narrow-minded and lacks a comprehensive picture. Some suggestions to improve the manuscript are as follows:

- (1) The authors made a segregation of 'spine synapse' and 'shaft synapse' based solely on the two-photon images in-vivo. However, caution shall be taken here, because the optical resolution under in-vivo imaging conditions like this cannot reliably tell apart whether a bright spot within or partially overlapping a segment of dendrite is a spine on top (or below) of it. Therefore, what the authors consider as a 'shaft synapse' (by detecting Ca²⁺ hotspots) has an unknown probability to be just a spine on top or below the dendrite. If there is other imaging data of higher axial resolution to validate or calibrate, the authors shall take some further considerations or analysis to check the consistency of their data, as the authors do need such a segregation between spine and shaft synapses to show how they evolve over the brain development stages.
- (2) The use of terminology 'bursts of spontaneous inputs' for describing voltage-clamp data seems improper. Conventionally, 'burst' refers to suprathreshold spike firing events, but here, the authors use 'burst' to refer to inward synaptic currents collected at the cell body. It is obvious that not every excitatory synaptic input (or ensemble of inputs) activation will lead to spike firing under naturalistic conditions, therefore, these two concepts are not equivalent. It is recommended to use 'barrage of inputs' instead of 'burst of inputs'. Imagine a full picture of the entire dendritic tree, the fact that the authors could always capture spontaneous Ca²⁺ events here and there within a few pieces of dendrites within an arbitrary field-of-view suggest that, the whole dendritic tree must have many more such events going on as a barrage while the author's patch electrode picks up the summed current flow from the whole dendritic tree.
- (3) Following the above issue, an analysis of the temporal correlation between synaptic (not segregating 'spine' or 'shaft') Ca²⁺ events and EPSCs is absent. Again, the authors drew arbitrary time windows to clump the events for statistical analysis. However, the demonstrated example data already show that the onset times of individual synaptic Ca²⁺ events do not necessarily align with the beginning of a 'barrage' inward current event.
- (4) The authors claim that "these observations indicate that the activity patterns investigated here are not or only slightly affected by low-level anesthesia". It would be nice to show some of the recordings in this work without any anesthesia to support this claim.
- (5) I suggest the authors should provide the number of cells and mice recorded in the figure legends.
- (6) Instead of showing only cartoon illustrations of dendrites in Figure 3-6, I suggest showing the two-photon images as well together with the cartoon.

The authors have addressed most of my issues, but I miss the responses to my points 5 and 6. I have no additional comments.

<https://doi.org/10.7554/eLife.93498.2.sa1>

Reviewer #3 (Public Review):

Summary:

There is a growing body of literature on the clustering of co-active synapses in adult mice, which has important implications for understanding dendritic integration and sensory processing more broadly. However, it has been unclear when this spatial organization of co-active synapses arises during development. In this manuscript, Leighton et al. investigate the emergence of spatially organized, co-active synapses on pyramidal dendrites in the mouse visual cortex before eye opening. They find that some dendrite segments contain highly active synapses that are co-active with their neighbors as early as postnatal day (P) 8-10, and that these domains of co-active synapses increase their coverage of the dendritic arbor by

P12-13. Interestingly, Leighton et al. demonstrate that synapses co-active with their neighbors are more likely to increase their activity across a single recording session, compared to synapses that are not co-active with their neighbors, suggesting local plasticity driven by coincident activity before eye opening.

The current manuscript includes some replication of earlier results from the same research group (Winnubst et al., 2015), including the presence of clustered, co-active synapses in the visual cortex of mouse pups, and the finding that synapses co-active with their neighbors show an increase in transmission frequency during a recording session. The main novelty in the current study compared to Winnubst et al. (2015) is the inclusion of younger animals (P8-13 in the current study compared to P10-15 in Winnubst et al., 2015). The current manuscript is the first demonstration that active synapses are clustered on specific dendrite segments as early as P8-10 in the mouse visual cortex, and the first to show the progression in active synapse distribution along the dendrite during the 2nd postnatal week. These results from visual cortex may help inform our understanding of sensory development more broadly.

Strengths:

The authors ask a novel question about the emergence of synaptic spatial organization, and they use well-chosen techniques that directly address their questions despite the challenging nature of these techniques. To capture both structural and functional information from dendrites simultaneously, the authors performed whole-cell voltage clamp to record synaptic currents arriving at the soma while imaging calcium influx at individual synaptic sites on dendrites. The simultaneous voltage clamp and calcium imaging allowed the authors to isolate individual synaptic inputs without their occlusion by widespread calcium influx from back-propagating action potentials. Achieving *in vivo* dendrite imaging in live mice that are as young as P8 is challenging, and the resulting data provides a unique view of synaptic activity along individual dendrites in the visual cortex at an early stage in development that is otherwise difficult to assess.

The authors provide convincing evidence that synapses are more likely to be co-active with their neighbors compared to synapses located farther away (Fig. 6F-H), and that synapses co-active with their neighbors increase their transmission frequency during a recording session (Fig. 7C). These findings are particularly interesting given that the recordings occur before eye opening, suggesting a relationship between co-activity and local synaptic plasticity even before the onset of detailed visual input. These results replicate previously published findings from P10-15 pups (Winnubst et al., 2015), increasing confidence in the reproducibility of the data.

The authors also provide novel data documenting for the first time spatially organized, co-active synapses in pups as young as P8. Comparing the younger (P8-10) and older (P12-13) pups, provides insight into how clusters of co-active synapses might emerge during development.

Weaknesses:

The P8-10 vs P12-13 age comparisons are the primary novel finding in this manuscript, and it is therefore critical to avoid systematic age differences in the methods and analysis whenever possible. In their rebuttal and revised manuscript the authors have acceptably addressed prior concerns regarding this important point, as well as most of the other methodological issues raised.

One point addressed in the rebuttal, but not corrected in the manuscript relates to the reliable localization of cells to visual cortex.

<https://doi.org/10.7554/eLife.93498.2.sa0>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:**Reviewer #1 (Public Review):***Summary:*

Using concurrent in vivo whole-cell patch clamp and dendritic calcium imaging, the authors characterized how functional synaptic inputs across dendritic arborizations of mouse primary visual cortex layer 2/3 neurons emerge during the second postnatal week. They were able to identify spatially and functionally separated domains of clustered synapses in these neurons even before eye-opening and characterize how the clustering changes from P8 to P13.

Strengths:

The work is technically challenging and the findings are novel. The results support previous EM and immunostaining studies but provide in vivo evidence on the time course and the trajectory of how functional synaptic input develops.

Weaknesses:

There are some missing details about how the experiments were performed, and I also have some questions about the analyses.

We have now added a more detailed description of the methods and added new supplemental figures and descriptions to clarify our analyses. Please find our responses to the specific points of this reviewer in the section “Recommendations for the authors” below.

Reviewer #2 (Public Review):

In this study, Leighton et al performed remarkable experiments by combining in-vivo patch-clamp recording with two-photon dendritic Ca²⁺ imaging. The voltage-clamp mode is a major improvement over the pioneer versions of this combinatorial experiment that has led to major breakthroughs in the neuroscience field for visualizing and understanding synaptic input activities in single cells in-vivo (sharp electrodes: Svoboda et al, Nature 1997, Helmchen et al, Nature Neurosci 1999; whole-cell current-clamp: Jia et al, Nature 2010, Chen et al, Nature 2011. I suggest that these papers would be cited). This is because in voltage-clamp mode, despite the full control of membrane voltage in-vivo not being realistic, is nevertheless most effective in preventing back-propagation action potentials, which would severely confound the measurement of individual synaptically-induced Ca²⁺ influx events. Furthermore, clamping the cell body at a strongly depolarized potential (here the authors did -30mV) also facilitates the detection of synaptically-induced Ca²⁺ influx. As a result, the authors successfully recorded high-quality Ca²⁺ imaging data that can be used for precise analysis. To date, even in view of the rapid progress of voltage-sensitive indicators and relevant imaging technologies in recent years, this very old 'art' of combining single-cell electrophysiology and two-photon imaging (ordinary, raster-scanned, video-rate imaging) of Ca²⁺ signals still enables measurements of the best level precision.

We thank the reviewer for reminding us of these important previous studies that we cite now in the revised manuscript.

On the other hand, the interpretation of data in this study is a bit narrow-minded and lacks a comprehensive picture. Some suggestions to improve the manuscript are as follows:

(1) The authors made a segregation of 'spine synapse' and 'shaft synapse' based solely on the two photon images in-vivo. However, caution shall be taken here, because the optical resolution under in vivo imaging conditions like this cannot reliably tell apart whether a bright spot within or partially overlapping a segment of the dendrite is a spine on top of (or below) it. Therefore, what the authors consider as a 'shaft synapse' (by detecting Ca²⁺ hotspots) has an unknown probability of being just a spine on top or below the dendrite. If there is other imaging data of higher axial resolution to validate or calibrate, the authors shall take some further considerations or analysis to check the consistency of their data, as the authors do need such a segregation between spine and shaft synapses to show how they evolve over the brain development stages.

We agree with the reviewer that the differentiation between spine and shaft synapses can be difficult for those spines that are located above or below the dendritic shaft in the z-dimension because of the lower resolution of 2-photon microscopy in the z-dimension compared to the image plane. We have now added a new paragraph to the Methods section to describe in more detail how we identify spine and shaft synapses and provide more examples in a new supplementary figure (Fig S5). We believe that we can identify spine and shaft synapses reliably in most cases, but added a cautionary note to make the reader aware of potential misidentifications.

(2) The use of terminology 'bursts of spontaneous inputs' for describing voltage-clamp data seems improper. Conventionally, 'burst' refers to suprathreshold spike firing events, but here, the authors use 'burst' to refer to inward synaptic currents collected at the cell body. Not every excitatory synaptic input (or ensemble of inputs) activation will lead to spike firing under naturalistic conditions, therefore, these two concepts are not equivalent. It is recommended to use 'barrage of inputs' instead of 'burst of inputs'. Imagine a full picture of the entire dendritic tree, the fact that the authors could always capture spontaneous Ca²⁺ events here and there within a few pieces of dendrites within an arbitrary field-of-view suggests that, the whole dendritic tree must have many more such events going on as a barrage while the author's patch electrode picks up the summed current flow from the whole dendritic tree.

We agree with the reviewer that “barrage” is a clearer term for multiple synaptic inputs occurring simultaneously and therefore we changed the terminology throughout the manuscript.

(3) Following the above issue, an analysis of the temporal correlation between synaptic (not segregating 'spine' or 'shaft') Ca²⁺ events and EPSCs is absent. Again, the authors drew arbitrary time windows to clump the events for statistical analysis. However, the demonstrated example data already shows that the onset times of individual synaptic Ca²⁺ events do not necessarily align with the beginning of a 'barrage' inward current event.

The reviewer writes that “an analysis of the temporal correlation between synaptic calcium events and EPSCs is absent”. We would like to point out that we did determine the percentage of calcium transients that occurred during barrages of synaptic inputs (~60%, page 7). This is important, since the barrages in our patch-clamp recordings most likely reflect spontaneous network events as described in the developing cortex previously by us and many other labs.

The time window we chose was not “arbitrary” as the reviewer suggests, but based on the duration of the barrages of synaptic inputs as defined in the Methods section.

The reason, why we did not perform a more in-depth analysis of the temporal relationship between synaptic calcium transients and synaptic input currents is that it is essentially impossible to relate calcium transients at individual synapses to specific synaptic input events. First, during barrages of synaptic inputs many synapses are active simultaneously, both in the mapped dendrites as well as in the un-observed parts of the dendritic arborization as the reviewer notes above. Thus, barrages cannot be broken down into individual synaptic transmission events. Second, since our acquisition frequency is ~10 Hz, we can identify the onset of individual synaptic calcium transients with 100-200 ms precision (1 or 2 frames). However, throughout any 100-200 ms period of recording, several synapses are active across the entire dendritic arborization such that we cannot assign a given calcium transient to a specific EPSC within a 100-200 ms epoch. Third, due to the limited clamping capacity of *in vivo* patch recordings, we cannot be certain that individual transmission events in distal dendrites can be resolved in the patch recording.

(4) The authors claim that "these observations indicate that the activity patterns investigated here are not or only slightly affected by low-level anesthesia". It would be nice to show some of the recordings in this work without any anesthesia to support this claim.

Indeed, the conclusion that the patterns of activity are only slightly affected by low levels of anesthesia is based on our previous recordings on the network level. Unfortunately, we are still not able to record calcium imaging with single synapse resolution in unanesthetized developing mice (and no one else is as far as we know), because the skull of these young animals is not firm, yet. As a consequence, movements cannot be reduced sufficiently for patching and imaging with single synapse resolution. Our previously published (Siegel et al., 2012) and unpublished work on the cellular level suggests that activity patterns during light anesthesia are very similar to those during sleep in mouse pups at this age.

Reviewer #3 (Public Review):

Summary:

There is a growing body of literature on the clustering of co-active synapses in adult mice, which has important implications for understanding dendritic integration and sensory processing more broadly. However, it has been unclear when this spatial organization of co-active synapses arises during development. In this manuscript, Leighton et al. investigate the emergence of spatially organized, coactive synapses on pyramidal dendrites in the mouse visual cortex before eye-opening. They find that some dendrite segments contain highly active synapses that are co-active with their neighbors as early as postnatal day (P) 8-10, and that these domains of co-active synapses increase their coverage of the dendritic arbor by P12-13. Interestingly, Leighton et al. demonstrate that synapses co-active with their neighbors are more likely to increase their activity across a single recording session, compared to synapses that are not co-active with their neighbors, suggesting local plasticity driven by coincident activity before eye-opening.

The current manuscript includes some replication of earlier results from the same research group (Winnubst et al., 2015), including the presence of clustered, co-active synapses in the visual cortex of mouse pups, and the finding that synapses co-active with their neighbors show an increase in transmission frequency during a recording session. The main novelty in the current study compared to Winnubst et al. (2015) is the inclusion of younger animals (P8-13 in the current study compared to P10-15 in Winnubst et al., 2015). The current manuscript is the first demonstration that active synapses are

clustered on specific dendrite segments as early as P8-10 in the mouse visual cortex, and the first to show the progression in active synapse distribution along the dendrite during the 2nd postnatal week. These results from the visual cortex may help inform our understanding of sensory development more broadly.

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The authors ask a novel question about the emergence of synaptic spatial organization, and they use well-chosen techniques that directly address their questions despite the challenging nature of these techniques. To capture both structural and functional information from dendrites simultaneously, the authors performed a whole-cell voltage clamp to record synaptic currents arriving at the soma while imaging calcium influx at individual synaptic sites on dendrites. The simultaneous voltage clamp and calcium imaging allowed the authors to isolate individual synaptic inputs without their occlusion by widespread calcium influx from back-propagating action potentials. Achieving in vivo dendrite imaging in live mice that are as young as P8 is challenging, and the resulting data provides a unique view of synaptic activity along individual dendrites in the visual cortex at an early stage in development that is otherwise difficult to assess.

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The authors also provide novel data documenting for the first time spatially organized, co-active synapses in pups as young as P8. Comparing the younger (P8-10) and older (P12-13) pups, provides insight into how clusters of co-active synapses might emerge during development.

Weaknesses:

This manuscript provides insufficient detail for assessing the rigor and reproducibility of the methods, particularly for age comparisons. The P8-10 vs P12-13 age comparisons are the primary novel finding in this manuscript, and it is, therefore, critical to avoid systematic age differences in the methods and analysis whenever possible. Specific concerns related to the age comparisons are listed below:

(1) Given that the same research group previously published P12-13 data (Winnubst et al., 2015), it is unclear whether both age groups in the current study were imaged/analyzed in parallel by the same researcher(s), or whether previous data was used for the P12-13 group.

While indeed the approach in the present study is similar to that of our previous study (Winnubst et al. 2015), the data set presented here is entirely new. The current study was made possible by a new microscope that allows combining resonant scanning with piezo-focusing to image large fractions of the dendritic arborization. In fact, we could now image almost 10 times larger dendritic segments including branch points than in our previous study. One author contributed to the experiments in both studies. Image analysis of all experiments was performed by the first author of the present study who was not involved in the Winnubst et al. work.

(2) The authors mention that they used 2 different microscopes, and used a fairly wide range of imaging frame rates (5-15 Hz). It is unclear from the current manuscript whether the same imaging parameters were used across the two age groups. If data for the two experimental groups was collected separately, perhaps at different times, by a different person, or on a different microscope, there is a concern that some differences between the groups may not necessarily be due to age.

The reviewer mentions that the experimental settings are not identical across the experiments of this study. In the original manuscript we erroneously reported in the Methods section that 2 different setups were used for this study; however, all experiments were performed on the same microscope. We have corrected this in the new manuscript. We took timelapse recordings of small stacks of varying depth to cover as many dendrites as possible in each recording, therefore, we needed to adjust the rate of acquired stacks within a certain range as the reviewer points out. The data were acquired by two scientists during an overlapping period. And while the different ages were not recorded in a strictly randomized fashion, they were not acquired in sequence according to ages, but rather involved many attempts on animals of different ages from many different litters. For each litter a small percentage of animals would generate successful recordings, and the ages of these successes were random. Therefore, we believe that neither the collection of data nor the analysis (see point above) affected the differences we describe here for the two age groups.

(3) It is unclear whether the image analysis was performed blind to age. Blinding to age during analysis is particularly important for this study, in which it was not possible to blind to age during imaging due to visible differences in size and developmental stage between younger and older pups.

The analysis was not setup to be performed blind to age. Not only is the age of the animal apparent at the stage (as the reviewer points out), also the number of spines and the activity levels clearly show differences between neurons only a few days apart. However, all age-related findings reported in this study - except the increase in synapse density and activity - became apparent to us only after the full set of synaptic transmission events was determined and the analysis was performed on the entire data set, making it very unlikely that event detection was biased.

(4) The relatively low N (where N is the number of dendrites or the number of mice) in this study is acceptable due to the challenging nature of the techniques used, but unintentional sampling bias is a concern. For example, if higher-order dendrites from the apical tuft were imaged at P12-13, while more segments of the apical trunk were imaged at P8-10, this could inadvertently create apparent age differences that were in fact due to dendrite location on the arbor or dendrite depth.

The reviewer points out that sampling bias with respect to synapse location along dendrites in the dataset could lead to falsely apparent age differences. In all experiments we imaged dendrites of layer 2/3 neurons that were relatively close to the cortical surface to optimize image quality. In addition, we confirmed that the mean distance of the imaged dendritic stretches from the cell body was similar between the dendrites of each age group (Young: 392 $\pm$ 104 μ m, Old: 323 $\pm$ 118 μ m; mean $\pm$ STD). Therefore, we do not think that sampling bias affected these results.

Additional general methodological concerns, which are not specifically related to the age comparisons, are listed below:

(5) The authors assert that clustered, co-active synapses emerge in the visual cortex before eye-opening, which is an important finding in that it suggests this phenomenon is driven by spontaneous activity rather than visual input. However, this finding hinges on the imaged cells being reliably located in the visual cortex, which is difficult to identify with certainty in animals that have not yet opened their eyes and therefore cannot undergo intrinsic signal imaging to demarcate the boundaries of the visual cortex. If the imaged cells were in, for example, nearby somatosensory cortex, then the observed spatial organization could be due to sensory input rather than spontaneous activity.

The reviewer argues that if the neurons included in our analysis were located in non-visual sensory cortex, e.g. the somatosensory cortex, sensory experience might have shaped clustered inputs instead of spontaneous activity. We are, however, certain that the neurons were located inside the primary visual cortex. In previous experiments where we performed the same craniotomies, we mapped spontaneous activity across the sensory areas in the occipital neocortex and we know the exact location of V1 which is already very consistent during the second postnatal week. (See for example Supplemental Figure 4 in Leighton et al., 2021).

(6) It is unclear how the authors defined a synaptic transmission event in the GCaMP signal (e.g. whether there was a quantitative $\Delta F/F$ threshold).

In the revised manuscript, we describe the procedure of identifying synaptic calcium transients in more detail and added a new supplemental figure to clarify this aspect of the analysis. In short, we use an automated detection with a 2x standard deviation threshold and a subsequent manual control and selection step. Please, find all details in the Methods section and Figure S4 of the revised manuscript.

(7) The authors' division of synapses into spine vs shaft is unconvincing due to the difficulty of identifying Z-projecting spines in images from 2-photon microscopy, where the Z resolution is insufficient to definitively identify Z-projecting spines, and the fact that spines in young animals may be thin and dim. The authors' examples of spine synapses (e.g. in Fig. 2A) are convincing, but some of the putative shaft synapses may in fact be on spines.

We agree with the reviewer that the differentiation between spine and shaft synapses can be difficult for those spines that are located above or below the dendritic shaft in the z-dimension because of the lower resolution of 2-photon microscopy in the z-dimension compared to the image plane (see also response to Reviewer 2, point 1). We have now added a new paragraph to the Methods section to describe in more detail how we identify spine and shaft synapses and provide more examples in a new supplementary figure (Fig S5). We believe that we can identify spine and shaft synapses reliably in most cases, but added a cautionary note to make the reader aware of potential misidentifications.

Reviewer #1 (Recommendations For The Authors):

I think the experiments performed were very technically challenging (probably one of the few labs that can do this in the field), and the findings provide in vivo evidence on how structured synaptic inputs are assembled during development that has never been reported.

I suggest improving the writing and presentation and really explaining how they conducted the experiments and how they defined shaft synapses.

Line 96: 12 dendritic areas from 11 mice at ages between postnatal day 8 to 13.

- Do the authors know how many neurons were imaged? It is unclear if the authors patch on all the imaged neurons and only imaged (or analyzed) the dendrites of those patched neurons. If yes, how sparse are the neurons labelled from IUE? From 1B, it looks like there are two cells adjacent to each other. Can the authors really distinguish whether the imaged dendrites are from the patched neuron?

The reviewer wonders whether we can tell apart dendrites of patched cells from those of neighboring neurons that were not patched. This is actually very straight forward: the experiment included a depolarization step (see Methods section) which leads to an immediate, but temporary, increase in fluorescence in all of the patched neurons' dendrites, but none of the neighboring dendrites. We have added this information to the Methods section of the new manuscript and provide now an example (Fig S3). Furthermore, as these cells normally fire frequently, it would immediately become clear that an unpatched cell is being imaged if backpropagating action potentials are predominantly observed rather than synaptic signals. The visualization of these synaptic signals is only possible due to the blockade of Na⁺ channels with QX314 in the intracellular solution (see Methods).

- In the methods section, it says 'dendrites were imaged in single plane or small stacks with plane...'. How do the authors do calcium imaging with small stacks of plane using Nikon MP scope?

Small stacks were acquired by using the piezo focusing device of our Nikon A1 microscope. Since we combined this fast focusing approach with resonant scanning, we were able to acquire z-stacks of 3-5 frames at a rate of up to 15 Hz (per stack).

- I also assume this is not chronic imaging, and there are different mice for each postnatal day. If it's true, this is somewhat important for all the correlation analysis as there are only 2 mice for each postnatal day (other than day 12) and day 13 only has 1 animal.

Yes, indeed these are not chronic experiments and dendrites imaged on different days are from different neurons and different mice. We agree with the reviewer that if it had been possible to image the same neurons across these developmental stages, we would have detected even clearer correlations. Therefore, we see our results as conservative estimates of the developmental trajectory of the analyzed parameters.

Line 104 - 109: I don't understand why the authors need to hold at -30mV to facilitate calcium influx through NMDA receptors? I assume this helps them to visualize as many synapses as possible? but wouldn't that also make the 'event frequency' not reflect the true value?

Indeed depolarizing the imaged neurons to -30 mV was necessary to get sufficient calcium influx to map synaptic inputs. We don't think that this affects the frequency of inputs, because the frequency of synaptic inputs is determined by the presynaptic firing rate and the release probability of the presynaptic terminal, which are not affected by the depolarization of the dendrite.

Figure 2A - It says in the method section that ROIs are manually selected. However, it's not explained what the criteria are. For spine synapses, it's easy to define but for shaft synapses like in Fig 2B, why are there 2 synapses on the shaft? And in Fig 4a, 5a, Fig S1 P13, some of the dendrites are packed with ROIs. What's the distance between those shaft synapses? Can the imaging resolution really separate them?

The reviewer asks for a better description of how we identified individual ROIs and thus synapse locations and whether this is actually feasible. We have now added a more detailed description of how we select synaptic sites based on the occurrence of synaptic calcium transients. In addition, we have added a new supplemental Figure (S4) to give the reader an impression of the image quality and the ability to locate individual synapses reliably. We find that separating sha synapses was possible for inter-synapse distances of $\sim 4 \mu\text{m}$ or more. The mean sha synapse distance in our data set is $21 \mu\text{m}$.

- Similar issue applies to Figure 4A that I'm not sure what's the resolution of each 'hot spot'. They all seem very close together. Maybe additional raw dendrite images with fluorescence changes like 1C or 2A could be helpful (or movies in the supplementary?)

As the reviewer suggests, we have added now additional supplemental figures to illustrate better how we identify synaptic transmission events as well as spine and sha synapses.

- Also for line 164, it says that 76% of high-activity synapses were located on spines. This could also maybe support that only the spine synapses are real synapses and many shaft synapses are actually not synapses and they were just categorized as shaft synapses from manual ROI?

We are actually quite sure that sha synapses are real synapses based on our analysis, since they show repeated synaptic calcium transients that co-occur with barrages of synaptic inputs as measured by patch-clamp recordings. Indeed one would expect to see a number of excitatory synapses on dendritic shas of pyramidal neurons at these ages based on previous EM studies (Miller and Peters, 1981; Wildenberg et al., 2023).

- While this might not impact the overall novelty of the paper, I would be curious to know if the authors can still observe the same findings if they only analyze spine synapses.

We repeated several analyses with a dataset that contained only spine synapses. For most analyses we observed the expected result: the effect sizes were similar compared to the entire data set, but the power was reduced. For example the effect of distance to closest high-activity neighbor and own activity (Fig 5E, F) was similar, but p-values were around 0.1 (Similar results for Figure 7B). In contrast, the co-activity with synapses within a domain was significantly higher than the co-activity with synapses in other domains also for the spine-synapse only data set.

Fig 6 - Does the domain co-activity also contribute to the synaptic current recorded (related to Fig 4).

Yes, the synaptic activity measured by calcium imaging contributes to the recorded EPSCs. However, the exact relationship between synaptic inputs measured by calcium imaging and those measured by patch-clamping is complicated by 3 factors: first, during barrages of synaptic inputs many synapses are active simultaneously, both in the mapped dendrites as well as in the un-observed parts of the dendritic arborization. Thus, barrages cannot be broken down into individual events. Second, since our acquisition frequency is $\sim 10 \text{ Hz}$, we can identify the onset of individual synaptic calcium transients with 100-200 ms precision (1 or 2 frames). However, throughout any 100-200 ms period of recording several synapses are active across the entire dendritic arborization such that we cannot assign a given calcium transient to a specific EPSC within a 100-200 ms epoch. Third, due to the limited clamping capacity of in vivo patch recordings, we cannot be certain that individual transmission events in distal dendrites can be resolved in the patch recording as EPSCs.

Reviewer #2 (Recommendations For The Authors):

(1) I suggest the authors should provide the number of cells and mice recorded in the figure legends.

The number of dendrites and mice is the same across all analyses: 12 dendrites from 11 mice for all experiments, 6/6 for P8-10 and 6/5 for P12-13. All dendrites and synapses (and their ages) are shown in the supplemental figures S1 and S2. We mention the number of imaged dendrites now at the beginning of the Results section and when we split ages for the first time.

(2) Instead of showing only cartoon illustrations of dendrites in Figures 3-6, I suggest showing the two photon images as well together with the cartoon.

The 2-photon images of all dendrites of the dataset are available in Figure S1. To allow the reader to compare the cartoon representations in the main figures and the 2-photon images of each neuron, we have now labeled each dendrite in the dataset (D1-D12, see figures S1 and S2). For every figure, where we show example neurons (cartoons or zoom ins) we now provide this identifier.

Reviewer #3 (Recommendations For The Authors):

To address the weaknesses outlined above, we recommend that the authors do the following:

- *To address concerns about the rigor and reproducibility of the methods specifically related to age comparisons, please confirm the following:*
- *Both age groups were run in parallel by the same researcher(s).*

Experiments were run partly overlapping and experiments from different age groups were performed in parallel by both researchers.

- *Both age groups were imaged on the same microscope, or animals from each age group were imaged on both microscopes. If it was necessary to use different microscopes for the different age groups for biological or practical reasons, please explain.*

All experiments were run on the same microscope, a Nikon A1 2-photon microscope. In the original methods description we erroneously mentioned two microscopes (copy and paste error from a previous publication). We corrected that in the revised manuscript.

- *There was no difference in imaging frame rates or other imaging parameters between age groups. If it was necessary to use different parameters for different age groups for biological reasons, please explain.*

We varied the frame rates somewhat to allow larger z-stacks for some experiments where dendrites traversed different depths; however the mean frame rates were similar between the experiments in P8-10 vs P12-13 dendrites, 8.5 vs 10 Hz, respectively.

- *Images were analyzed blind to age.*

The analysis was not setup to be performed blind to age. The number of spines and the activity levels clearly show obvious differences between neurons only a few days apart. However, all findings reported in this study related to age - except the increase in synapse

density and activity - became apparent to us only after the full set of synaptic transmission events was determined and the analysis was performed on the entire data set, making it unlikely that event detection was biased.

- There was no difference in the location of analyzed dendrites (e.g. depth from the pia, branch order) between age groups.

In all experiments we imaged dendrites of layer 2/3 neurons that were relatively close to the cortical surface to optimize image quality. In addition, we determined the mean distance of the imaged dendritic stretches from the cell body and found that this distance was similar between the dendrites of each age group (Young: 392 +/- 104 μ m, Old: 323 +/- 118 μ m; mean +/- STD). Therefore, we do not think that sampling bias affected these results.

• To address general methodological concerns, please provide additional description of the following points:

- Please clarify how the visual cortex was identified in P8-13 pups. If there was ambiguity about identifying the visual cortex in these pups, please discuss the implications of this ambiguity.

The reviewer asks how we identified V1 in these experiments. We are indeed certain that the neurons were located inside the primary visual cortex. We have ample experience with mapping V1 in these animals based on patterns of spontaneous activity as well as post-hoc stainings. V1 is quite large already at these ages (> 2 mm long and > 1 mm wide) and its extent very consistent across animals. Thus, we would argue it is actually hard to miss.

- Please clarify how synaptic transmission events were identified in the GCaMP signal.

We have now added a more detailed description of how we identify synaptic calcium transients. In addition, we have added a new supplemental Figure (S3) to give the reader an impression of the image quality and the ability to locate individual synapses reliably.

- It is acceptable to use the spine vs shaft analysis despite the inevitable difficulty resolving Z-projecting spines, but this caveat should be mentioned in the discussion of the spine vs shaft results.

We added a more detailed description of spine and shaft synapse identification, a new supplemental figure (S5) and we now mention the caveat related to the limited z-resolution of 2-photon microscopy in the revised manuscript.

• Two additional minor details should be clarified in the text of the manuscript:

- Please specify the volume of DNA solution injected into each embryo.

The injected volume was 1 μ l. We added this information in the Methods section of the revised manuscript.

- In Fig S1, please specify whether the scale bar applies to all images.

The scale bar applies to all images. This information was added to the figure legend.

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<https://doi.org/10.7554/eLife.93498.2.sa4>